Model Analysis Report

Event Wind Lookup

MKM-WS-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Symmetric radial profile $V_{sym}(r)$ in m/s by distance from the storm centre (R_{max} = 30 km, R_{outer} = 250 km).

r (km)	V_{sym} at $V_{max} = 35$	V_{sym} at $V_{max} = 50$	V_{sym} at $V_{max} = 65$
0	14.0	20.0	26.0
10	21.0	30.0	39.0
20	28.0	40.0	52.0
30	35.0	50.0	65.0
50	34.3	48.6	62.7
75	32.8	45.4	57.6
100	30.9	41.4	51.4
150	26.5	32.8	38.3
200	21.9	24.5	26.7
250	17.5	17.5	17.5
300	13.6	12.0	10.9
400	7.72	5.06	3.72

Table 2: Outer-decay length L (km) by peak wind, anchored at $V_{outer,ref}$ = 17.5 m/s with $p = 1.5$. The geometric fallback applies when the eyewall wind is at or below the anchor.

V_{max} (m/s)	L (km)	Fallback
15.0	220.0	yes
17.5	220.0	yes
20.0	842.1	no
30.0	332.2	no
40.0	249.8	no
50.0	213.0	no
60.0	191.4	no
70.0	176.9	no

Table 3: Asymmetry multiplier $1+\epsilon\cos(\theta-\phi)$ by translation speed and azimuth ($V_{max} = 50$ m/s, heading 315°, $\phi = 45°$, $\epsilon_{max} = 0.3$).

u (km/h)	ϵ	$\theta = 0°$	$\theta = 45°$	$\theta = 90°$	$\theta = 135°$	$\theta = 180°$	$\theta = 270°$
0.0000	0.0000	1.00	1.00	1.00	1.00	1.00	1.00
5.00	0.0163	1.01	1.02	1.01	1.00	0.9884	0.9884
10.0	0.0327	1.02	1.03	1.02	1.00	0.9769	0.9769
18.0	0.0588	1.04	1.06	1.04	1.00	0.9584	0.9584
25.0	0.0817	1.06	1.08	1.06	1.00	0.9422	0.9422
35.0	0.1144	1.08	1.11	1.08	1.00	0.9191	0.9191
50.0	0.1634	1.12	1.16	1.12	1.00	0.8845	0.8845

Table 4: End-to-end point wind (m/s) due north of the storm centre, over sea vs over land ($\rho_{sea} = 1.0$, $\rho_{land} = 0.8$).

Distance (km)	Sea (m/s)	Land (m/s)	Difference
0	20.8	16.7	-4.17
25	46.9	37.5	-9.37
50	50.6	40.5	-10.1
100	43.1	34.5	-8.63
150	34.1	27.3	-6.82
250	18.2	14.6	-3.65

Table 5: Linear state interpolation between stored hours 12 and 18. Queries outside the trajectory range clamp to the first / last stored state.

Hour	V_{max} (m/s)	R_{max} (km)	Clamped
6.00	40.0	40.0	yes
12.0	40.0	40.0	no
13.5	45.0	36.2	no
15.0	50.0	32.5	no
16.5	55.0	28.8	no
18.0	60.0	25.0	no
24.0	60.0	25.0	yes

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 6: Stress testing scenarios for Event Wind Lookup.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	$+2\sigma$ inputs	Medium	Pending
Severe stress	$+3\sigma$ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 7: Performance metrics for Event Wind Lookup.

Metric	**Value**	**Status**
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.